INTERNSHIP · PROCESS ENGINEERING

Scrap analysis on automated cut-and-crimp machines

DRÄXLMAIER, Zeta production zone · Sousse · Aug – Sep 2026

3
Komax 633 machines studied (588, 590, 596)

5.3–8 kg
weighed scrap per day from the 3 machines

23 / 47
logged faults were the crimp-feed fault (CFA)

CONTEXT

DRÄXLMAIER's Zeta zone cuts, strips, crimps and inserts wires for automotive harnesses. Scrap there was a company priority, and my job was to find where it came from.

WHAT I DID

- Weighed the scrap of each machine separately and compared it with the zone scrap rate (g/min) from the MES.
- Ranked the machine fault log with a Pareto chart; the crimp-feed fault came first.
- Traced causes with a 5M Ishikawa diagram, then scored them with AMDEC (severity × frequency × detectability).
- Worn blades, press-tool settings and cable-feed parameters came out as the most critical causes (scores 24, 20 and 24).

TOOLS AND METHODS

QQOQCP · Pareto · Ishikawa · AMDEC · MES data · Root-cause analysis

OUTCOME

Proposed shorter preventive-maintenance intervals, a visual check before each start-up and recurring first-level maintenance training for operators. Scrap decreased in August after partial implementation; a longer follow-up is needed to quantify the gain.

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